

Yusuf Brima

Contact

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Summary

Highly motivated AI/ML Researcher with over 6 years of experience in digital signal processing, machine learning, and deep learning. Proficient in Python, PyTorch, and TensorFlow, with a strong track record of impactful research contributions in top-tier conferences and journals. Passionate about interdisciplinary research.

Core skills

Programming languages: Python, R, C/C++, SQL, JavaScript, HTML/CSS

Machine Learning & Data Science: TensorFlow, Keras, PyTorch, SciKit-Learn, CUDA, NumPy, Pandas, Matplotlib, Seaborn, Plotly, Dash

Systems: HPC with SLURM, Grid Computing with Oracle SGE, Docker, GCP, GPU Computing

Image Processing: OpenCV, Scikit-Image, SimpleITK, NiBabel, Pydicom, ANTsPy

Explainable & trustworthy AI: SHAP, Captum, Fairlearn, AIF360, LIME, TorchCAM

Databases: MySQL, PostgreSQL, Oracle, SQL Server

Software and Productivity: Git/GitHub, LaTeX, Bash, Google Workspace, Microsoft Office, Jira

Operating Systems: Linux, Unix, Windows

Languages: English (native-level), German (intermediate), French (basic)

Education

- **Universität Osnabrück, Germany** *Nov 2021 – Dec 2025*
 - Ph.D. Cognitive Science
 - Award: DFG Research Grant for early-stage researchers.
- **African Institute for Mathematical Sciences (AIMS), Rwanda** *Sept 2020 – Jul 2021*
 - MSc. Mathematical Sciences.
 - Award: AIMS & MasterCard Foundation Scholarship.
- **University of Dhaka, Bangladesh** *Jul 2018 – Aug 2020*
 - MSc. Computer Science & Engineering.
 - Award: Queen Elizabeth Commonwealth Scholarship.
- **University of Makeni, Sierra Leone** *Sept 2012 – Feb 2017*
 - BSc. Computer Science.
 - Award: Undergraduate Merit Scholarship.

Professional Experience

- **Osnabrück University, Germany – Biomedical AI Fellow** (Jan 2025–present)
 - Conduct research in multimodal representation learning for biomedical data, focusing on early disease diagnosis and prognosis, with emphasis on robustness, explainability, fairness, and generalization.
 - Co-supervise students on projects in medical imaging and clinical AI, including transformer-based 3D medical image analysis and explainable/fair AI for low-resource healthcare settings.
 - Participate in conferences, workshops, and seminars; publish research output in top-tier conferences and journals including MICCAI, npj Digital Medical, IEEE, etc.
- **University of Osnabrück, Germany – Research Associate** (Nov 2021–Dec 2024)
 - Conducted research focused on deep representation learning, including self-supervised learning methods such as contrastive, generative, and joint-embedding architectures, aiming toward representation robustness, generalization, transferability, and interpretability via

- disentanglement.
- Improved data processing efficiency by at least 20% and model accuracy by at least 25% on target tasks.
- Curated the SynSpeech dataset and maintained open-source access.
- Developed and maintained open-source code implementations for all published research, available on GitHub
- (Co-)organized presentations and seminars to showcase research at venues including NeurIPS (2023), German Conference on Artificial Intelligence Doctoral Consortium (2023), and ICLR (2024).
- Mentored and supervised 5+ students thesis, study projects and term papers.
- **University of Makeni, Sierra Leone – Research Teaching Assistant** (Feb 2017–Jul 2018)
 - Taught computer science courses to 40+ students; supervised 4+ dissertations; achieved 90% project completion rate.
 - Developed course materials, homework assignments, and exams. Guided students’ thesis work.
- **Limkokwing University of Creative Technology, Sierra Leone – Lecturer** (Jan 2018–Jul 2018)

Taught courses on programming logic and design for approximately 120 students, in classes of approximately 40 students.

Selected publications

Lead author

- Brima Y, Atemkeng M. An Multimodal Explainable Deep Learning Framework for Alzheimer’s Disease Diagnosis using 3D Magnetic Resonance Imaging and Clinical Data. Submitted to *npj Digital Medicine*.
- Brima Y, Atemkeng M, Kallon LH, Niyukuri D, Vacavant A, Saidu S, Chen DG. Few-shot Cross-country Generalization of Tabular Machine Learning and Foundation Models for Childhood Anemia Prediction under Distribution Shift. arXiv preprint arXiv:2605.26589, 2026. Under review at *BioData Mining*.
- Brima Y, Atemkeng M, Ngueajio M, et al. Artificial Intelligence in Neglected Tropical Diseases: Current Applications, Challenges, and Opportunities. Accepted at *npj Digital Medicine*.
- Brima Y, Atemkeng M. An empirical study of machine learning robustness and scalability for imbalanced tabular clinical data in emergency and critical care. *Scientific Reports* 16, 18004 (2026).
- Brima Y, Krumnack U, Pika S, Heidemann G. Understanding Self-Supervised Learning of Speech Representation via Invariance and Redundancy Reduction. *Information*. 2024 Feb 15;15(2):114.
- Brima Y, Krumnack U, Pika S, Heidemann G. Learning Disentangled Audio Representations through Controlled Synthesis. Presentation at ICLR Tiny Papers Track 2024. arXiv:2402.10547.
- Brima Y, Atemkeng M. Saliency-driven Explainable Deep Learning in Medical Imaging: Bridging Visual Explainability and Statistical Quantitative Analysis. *BioData Mining*, 2024, 17(1):18.
- Brima Y, Krumnack U, Pika S, Heidemann G. Learning Disentangled Speech Representations. *New in Machine Learning Workshop, NeurIPS 2023*, New Orleans, USA.
- Brima Y, Atemkeng M, Tankio Djiokap S, Ebiele J, Tchakounté F. Transfer Learning for the Detection and Diagnosis of Types of Pneumonia Including COVID-19 from Chest X-ray Images. *Diagnostics*, 2021, 11(8):1480.
- Brima Y, Kamal Tushar MH, Kabir U, Islam T. Deep Transfer Learning for Brain Magnetic Resonance Image Multi-class Classification. *Dhaka University Journal of Applied Science & Engineering*, 2022.

Co-authored

- Kamga LS, Tchoupe Tchendji M, Atemkeng M, **Brima Y**, Bilal H, Lalléchère S, Rossignol J, Ravelo B. Interpretable Slice-Based Alzheimer’s Disease Classification: A Class Activation Mapping Analysis of Entropy-Guided 2D Sampling. Under review at PACAIS 2026.
- Kamga LS, Tchoupe Tchendji M, Atemkeng M, Brima Y, Bilal H, Lalléchère S, Rossignol J,

Ravelo B. Task-Aware 2D Sampling via Prediction Entropy for Multi-Axis MRI Classification of Alzheimer's Disease. Under review at *Cognitive Computation*.

- Nguezet PV, Fute ET, **Brima Y**, Azanguezet BM, Atemkeng M. From Visual Saliency to Anatomical Explanation: Generating Textual Diagnostic Reports from Brain MRIs with Saliency-Grounded LLMs. Under review at *BioData Mining*.
- Nhlapho W, Atemkeng M, **Brima Y**, et al. Transformers for 3D medical image analysis: a systematic review of architectural innovations, performance, and clinical applications. *Artificial Intelligence Review* (2026).
- Singh D, *Brima Y*, Levin F, Becker M, Hiller B, Hermann A, Villar-Munoz I, Beichert L, Bernhardt A, Buerger K, Butryn M. An unsupervised XAI framework for dementia detection with context enrichment. *Scientific Reports*, 2025, 15(1):39554.
- Hamlomo S, Atemkeng M, **Brima Y**, Nunhokee C, Baxter J. A Systematic Review of Low-rank and Local Low-rank Matrix Approximation in Big Data Medical Imaging. *Neural Computing and Applications*, 2025.
- Nhlapho W, **Brima Y**, Atemkeng M, Ndogmo J-C. Bridging the gap: exploring interpretability in deep learning models for brain tumor detection and diagnosis from MRI images. *Information*, 2024, 15(4):182.

Further education

- AI for Medicine Specialization, DeepLearning.AI (2026)
- AI for Healthcare Specialization, Stanford (2025)
- Deep Learning Bootcamp, MUST, North-West University, South Africa (2025)
- Confidentiality and Data Protection in Health Research, CITI (2025)
- AI Engineering Professional Certificate, IBM (2023)
- Data Science Professional Certificate, IBM (2023)
- Python Programming and Data Structures, Coursera, University of Michigan (2023)
- Mediterranean Machine Learning School, Università Milano Bicocca, Italy (2022)
- Deep Learning Program, Neuromatch Academy (2022)

Selected grant & scholarships

- Google Cloud Academic Research Grant (2026)
- CIFAR Deep Learning and Reinforcement Learning (DLRL) Summer School Inclusive AI, Fully-funded Scholarship, Canada (2024)
- Deep Learning Indaba, Fully-funded Scholarship (2023)
- Cambridge Ellis Machine Learning Summer School, Fully-funded Scholarship (2022)

Academic engagement & memberships

- Member of the Master's Program in Mathematical Sciences in Mathematical Epidemiology Scholarship Selection Committee, African Institute for Mathematical Sciences and the Africa Health Collaborative (2026)
- Member of the Deep Learning Indaba Kambule Doctoral Award Committee (2025)
- Co-organizer, "Bridging Biological and Artificial Neural Networks" workshop, University of Osnabrück (2022)
- Co-organizer, Deep Representation Learning Seminar, University of Osnabrück (2023/24)
- Peer reviewer: PA-AISS 2026 Program Committee (PC), Scientific Reports, BMC Medical Informatics and Decision Making, Healthcare Technology Letters, npj journals, ICLR, ICML, MusIML Workshop, Deep Learning Indaba (since 2023)
- Approved access to Demographic and Health Surveys (DHS) data for research on reliable disease diagnosis (2026)
- Access authorization and training for controlled biomedical datasets and platforms, including UK Biobank/MRC, PhysioNet, LONI IDA, and the German FDPG framework (2025)
- Memberships: MICCAI Society (since 2023), Black in AI (since 2022), Deep Learning Indaba (since 2022), IEEE (since 2019), AIMS Alumni Network (since 2021)